

FARM ENVIRONMENT PLAN



18108



L M FARMS

PLAN DEVELOPED WITH:

ABOUT YOUR FARM ENVIRONMENT PLAN

This plan is tailored to your farm – it outlines your current environmental practices, and any actions or management needed to help improve soil health, water quality, biodiversity, animal wellbeing and emissions reduction.

The actions in this plan follow nationally approved Good Farming Practices, to help you meet any regional council or central government requirements.

HOW TO USE THIS PLAN

This plan prioritises focus areas to protect your farm for future generations. We're here to help you tick things off the list or manage any changing conditions on farm - just reach out to your Sustainable Dairying Advisor.

You can also give us a call 24/7 on 0800 65 65 68.

**Keep track of actions through the Dairy Diary app
Find the app on the Google Playstore or Apple Store,
or access your actions through your farm source
website.**

APP STORE



GOOGLE PLAY



MORE INFO



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YOUR FARM DETAILS

SUPPLIER NUMBER 18108

FARM OWNER

L M Farms

FARM OWNER ADDRESS

9 Smith-Grey Crescent
Hikurangi 0184

FARM ADDRESS

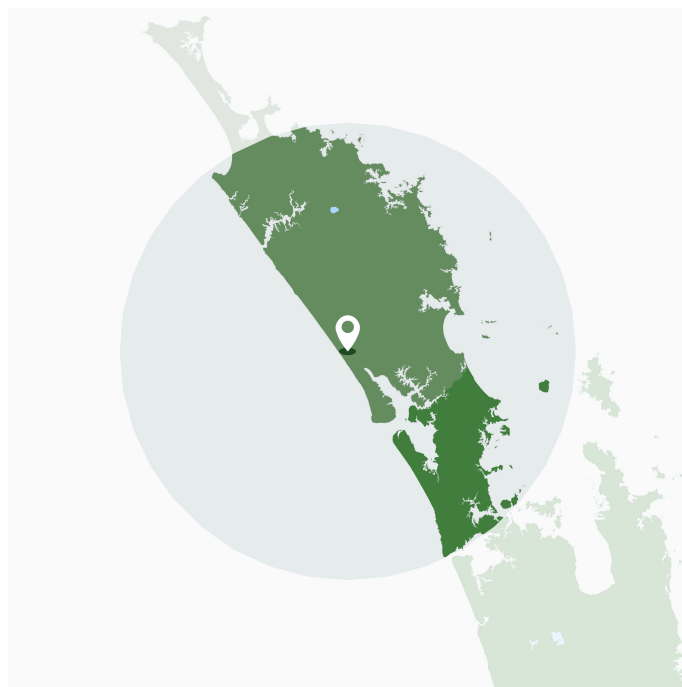
West Coast Road
Dargaville

REGIONAL COUNCIL

Northland

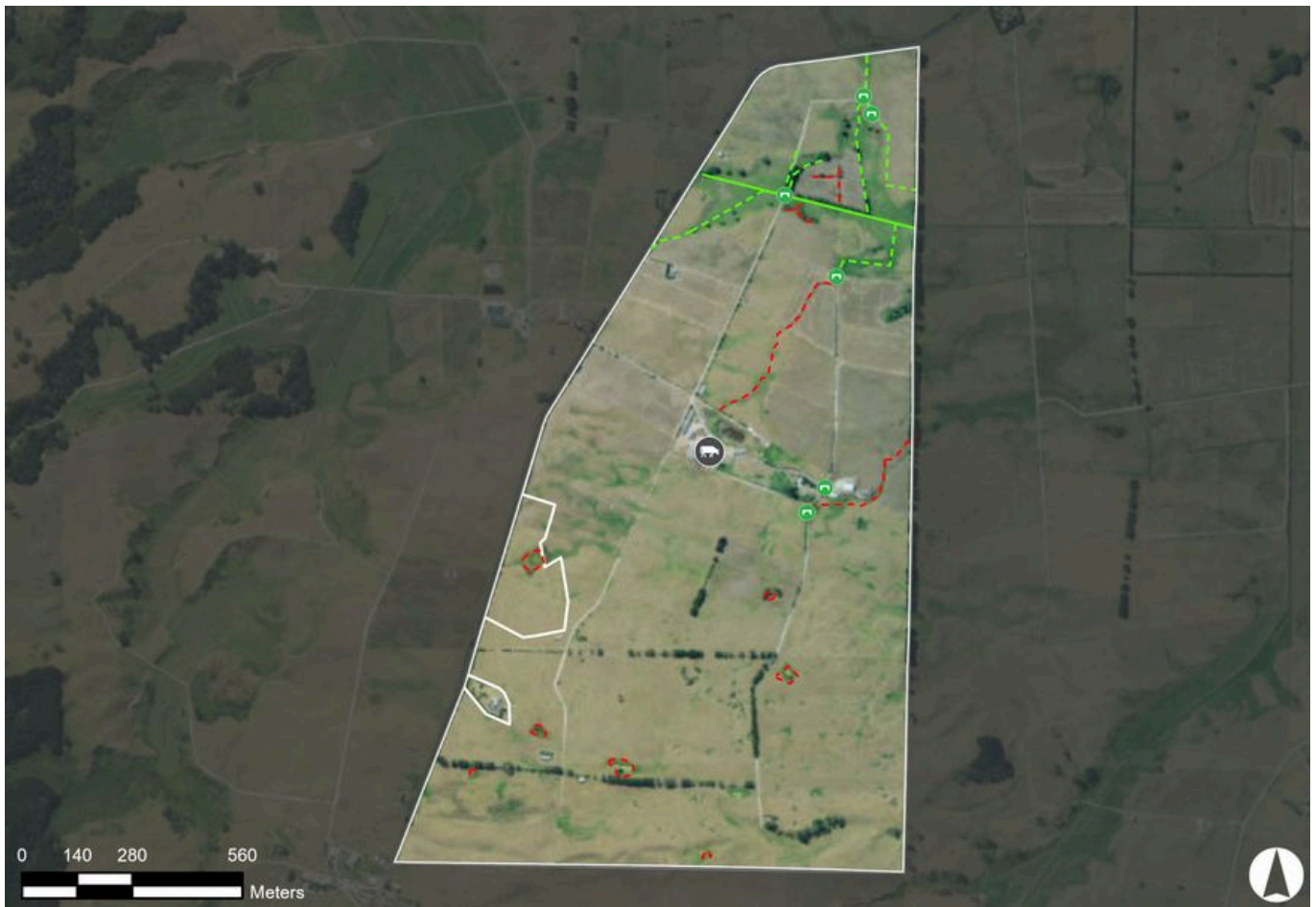
PLAN LAST EDITED

14 December 2025



FARM OVERVIEW MAP

The map below presents the land in which the farming operations covered in this document occur and identifies some key points of interest. More detailed maps looking at specific environmental management topics are contained throughout the document.



Major Stock Excluded Waterway



Minor Stock Excluded Waterway



Minor Stock Not Excluded Waterway



Compliant Crossing



Main Dairy Platform



Support Block



Dairy Shed

GOOD FARMING PRACTICES

FARM MANAGEMENT

The characteristics of the farm and the farm system are identified		ACHIEVED
A risk assessment of the farms inherent and management activity risks is undertaken		ACHIEVED
Accurate and auditable records are kept of annual farm inputs, outputs and management practices		ACHIEVED
Fertiliser is stored and loaded to minimise the risk of spillage and losses to waterways and groundwater		ACHIEVED
Feed is stored, transported and fed to minimise wastage, leachate and soil damage		ACHIEVED
Hazardous substances (fertilisers, agrichemicals and fuel) are stored, handled, used and disposed of to avoid contamination of waterways and groundwater		ACHIEVED
Farm waste is minimised		ACHIEVED

NUTRIENT MANAGEMENT

Soil phosphorus levels are monitored and maintained below or within the target ranges for the soil-type and crop		ACHIEVED
The amount and timing of fertiliser inputs, takes account of all sources of nitrogen and phosphorus, matches plant requirements and minimises losses to waterways and groundwater		ACHIEVED
Fertiliser spreading equipment is maintained and calibrated		ACHIEVED

WATER USE & IRRIGATION MANAGEMENT

Dairy shed and stock water use is efficient and prevents source contamination		ACHIEVED
The depth, rate and timing of irrigation is managed to meet plant demand and minimise the risk of leaching and run-off		N/A
The irrigation system is designed, operated and regularly checked to minimise the amount of water applied to meet plant demand, and prevent microbial contamination		N/A

WATERWAYS & BIODIVERSITY

Stock are excluded from lakes and waterways

ACHIEVED

Farm indigenous biodiversity and Mahinga Kai values are identified and protected



ACHIEVED

EFFLUENT MANAGEMENT

Effluent and manure is applied at depths, rates and amounts that match plant requirements and minimise the loss of nutrients or microbial pathogens to waterways and groundwater



ACHIEVED

The effluent system is designed, operated and regularly checked to minimise the risk of nutrient and microbial pathogen loss to waterways and groundwater, and to prevent microbial contamination



ACHIEVED

LAND & SOIL MANAGEMENT

Critical Source Areas and farm Hot Spots are identified and managed to minimise contaminant losses to waterways

ACHIEVED

Cultivation is managed to reduce the risk of sediment loss and maintain soil structure



ACHIEVED

Erosion-prone land is managed or retired to minimise soil losses



ACHIEVED

Grazing of pastures and crops is managed to minimise sediment and contaminant loss

ACHIEVED

Paddocks selected for Intensive Winter Grazing (including intensive baleage wintering) are low risk and managed to minimise the risk of erosion, run-off to waterways and leaching to groundwater

ACHIEVED

GREENHOUSE GAS EMISSIONS

Farm greenhouse gas emissions are known and a plan is in place to reduce or offset them, that also considers adaptation to climate change



ACHIEVED



Indicates the achievement of the Good Farming Practice will also impact Greenhouse Gas Emissions.

SUMMARY OF YOUR ACTIONS

This list includes all actions and recommendations that have been agreed as part of this Farm Environment Plan. Actions are required to achieve Good Farming Practices. Actions that have a target date within 2 years are captured as “Current Actions”. Actions with a target set more than 2 years in the future are captured as “Future Actions”. “Recommendations” cover all other actions that are Leading Practice actions (beyond GFP) or are actions which are not related to a GFP.

Please review the Good Farming Practices in each section of your FEP. These identify the actions or practices you are already doing and need to continue doing to meet the Good Farming Practice standards.

KEY: ACTION PRIORITY

 Low

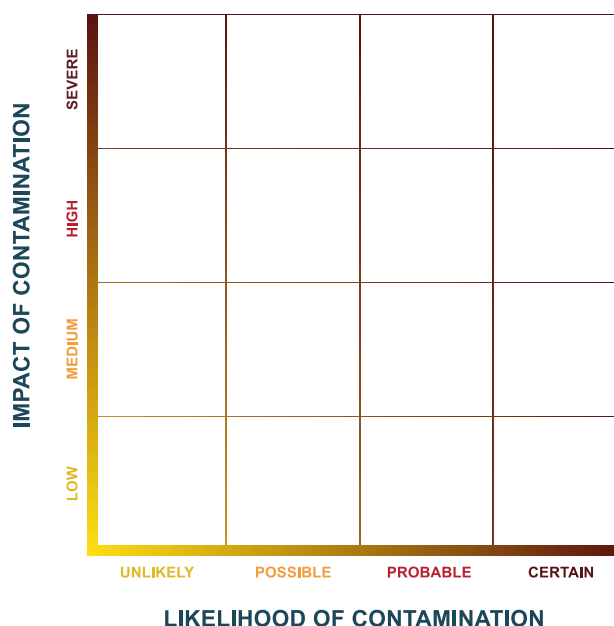
 Medium

 High

 Critical

UNDERSTANDING THE RISKS ON YOUR FARM

This section provides some context to help understand the relative impact and likelihood of environmental risks that have been identified on your farm. The chart on this page together with the map on the following page can be useful when thinking about what environmental risk areas on your farm need the most focus.



HOW ARE RISK RATINGS MEASURED?

The issues plotted on the chart above have been done so based upon two measures that are assigned to a specific area of your farm where an environmental risk has been identified. 1. Impact of contamination (on the vertical axis, or the first dial) is a measure of the potential scale or significance of contaminants that may be lost from this area of your farm. It's about quantifying how bad could the outcome for the environment be; 2. Likelihood of contamination (on the horizontal axis, or the second dial) is about the chance of the contamination actually occurring from that area of your farm. It takes into account things like how far the area might be from waterways as well as the slope or aspect of the area; When combined together the two measures also give an overall 'risk rating'. The measures and the combined rating are presented for each risk area along with other descriptive information about the risk area on the subsequent pages of this document.

EXAMPLE



UNDERSTANDING THE RISKS ON YOUR FARM

The map below shows the location of the risk areas identified on your farm. The Risk Rating presented here is a combined measure of the impact and likelihood of contamination occurring from each risk area.



Low



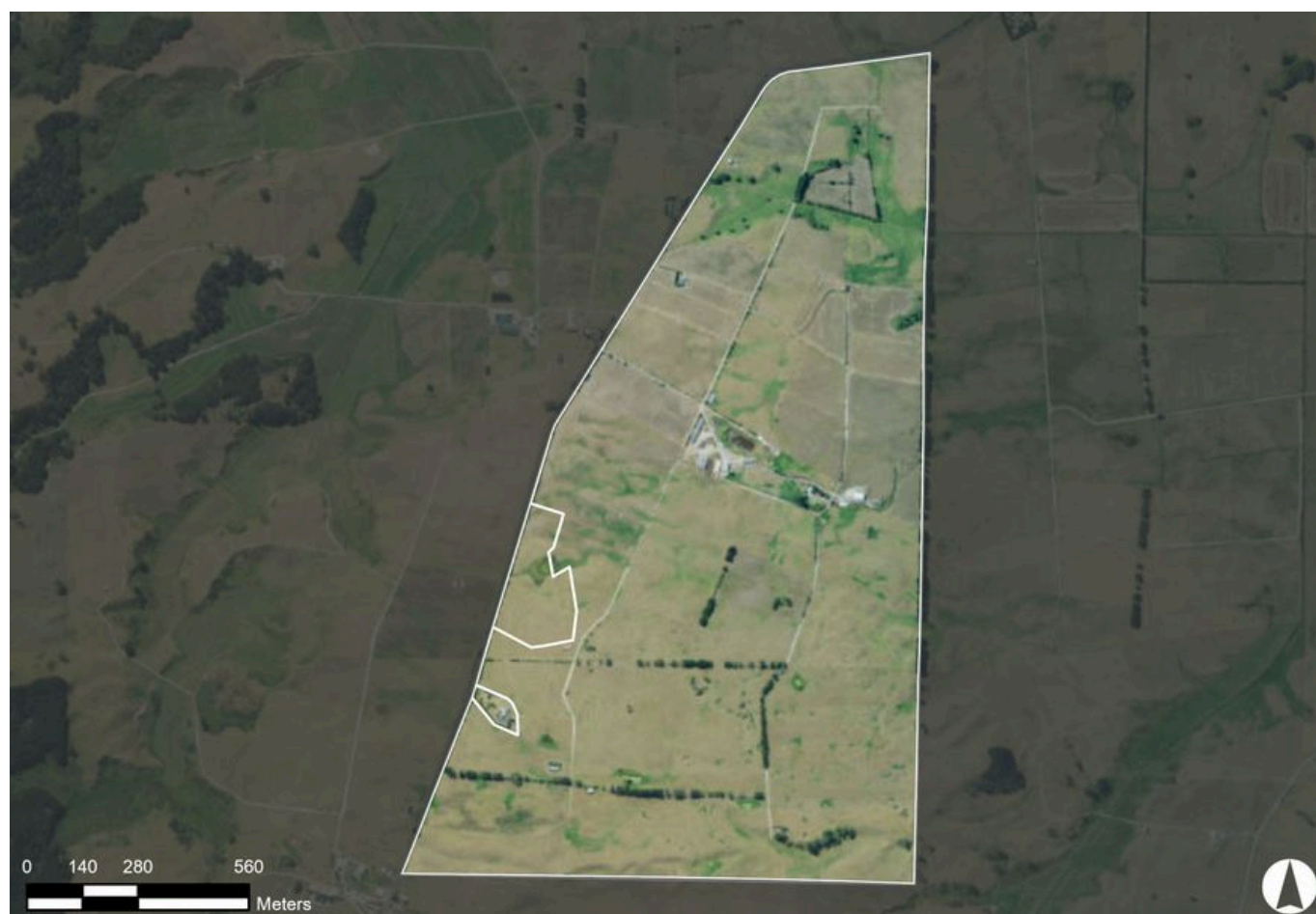
Medium



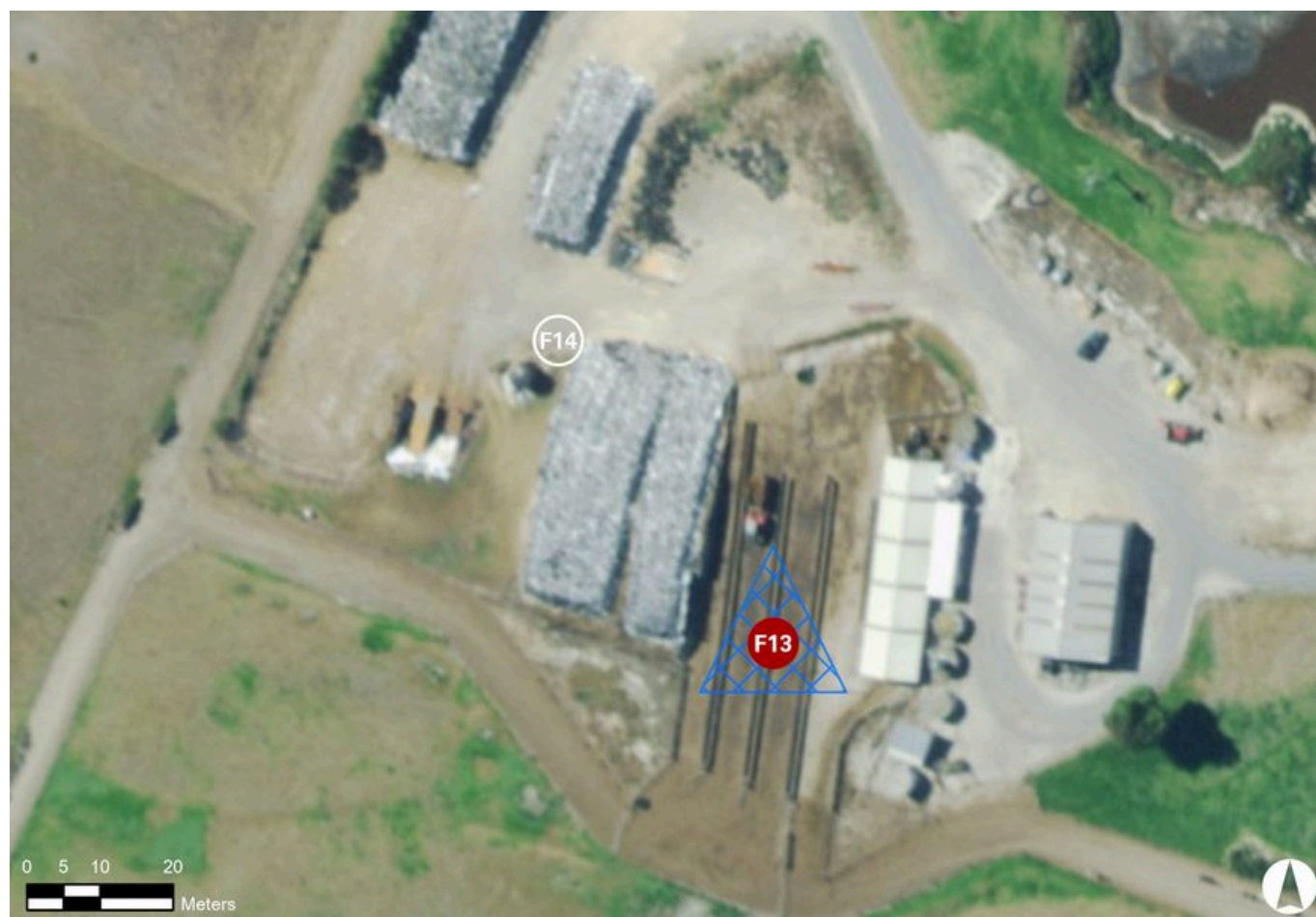
High



Critical



FARM MANAGEMENT MAHI WHAKAHAERE



F13 Feed pad

F14 Feed storage area

FARM MANAGEMENT

GOOD FARMING PRACTICES

The characteristics of the farm and the farm system are identified



ACHIEVED

A risk assessment of the farms inherent and management activity risks is undertaken



ACHIEVED

Practices:

Risk factors to water quality associated with the landscape and farm system have been assessed and are managed appropriately

Accurate and auditable records are kept of annual farm inputs, outputs and management practices



ACHIEVED

Practices:

Accurate and auditable records of annual farm inputs, outputs and management practices are maintained that support the actions being undertaken to achieve the Dairy Good Farming Practices and reduce any additional risks identified through the risk assessment

Fertiliser is stored and loaded to minimise the risk of spillage and losses to waterways and groundwater



ACHIEVED

Practices:

The Fertiliser Industry - Code of Practice for Fertiliser handling, storage and use is followed

Fertiliser storage sites are:

- Located away from waterways or areas prone to flooding
- Well ventilated with adequate lighting
- Appropriately signed
- Able to contain a spillage and provide secondary containment where appropriate

Stored fertiliser is covered

Feed is stored, transported and fed to minimise wastage, leachate and soil damage



ACHIEVED

Practices:

Rainfall run-off is diverted to land away from feed storage areas

Silage is sufficiently wilted before being put into stack

Silage remains sealed while stored to prevent rotting

Permanent feed-out areas / facilities are sealed and all run-off is collected and applied to land via the effluent system

Feed-out areas are located away from critical source areas

FARM MANAGEMENT

GOOD FARMING PRACTICES CONTINUED

Hazardous substances (fertilisers, agrichemicals and fuel) are stored, handled, used and disposed of to avoid contamination of waterways and groundwater

ACHIEVED

Practices:

An inventory of all hazardous substances stored on-farm is kept, including Safety Data Sheets (SDS)

All hazards are identified, and staff made aware of these and how they are to be managed

Hazardous substance information and training is provided

A Certified Handler certificate is held if Class 6.1A or 6.1 B are stored or used on site by farm staff

Appropriate Personal Protective Equipment is made available, well-maintained, and worn

Procedures are in place for managing emergencies

Fertilisers, agrichemicals, and fuels are stored separately

Applications follow the Safety Data Sheet (SDS) conditions and are only when weather conditions are suitable

Re-entry and withholding periods are adhered to

Storage locations are:

- Located away from waterways or areas prone to flooding
- Well ventilated with adequate lighting
- Appropriately signed
- Able to contain a spillage and provide secondary containment where appropriate
- Agrichemicals are stored in containers constructed of non-flammable material

Farm waste is minimised

ACHIEVED

Practices:

All inorganic, non-recyclable waste is contained and removed from farm

Dead animals are sent off farm for processing or correctly disposed on-farm

Pests are controlled around feed storage and waste infrastructure

FARM MANAGEMENT

F4

FARM OVERVIEW

The dairy farm is currently operating under the entity L M Farms. The day to day farming operation is actively managed by Lana Gillett and Michael Slagter (Farm owners). The owners have lived and farmed this land for a number of years therefore have an in-depth understand the lands advantages and its limitations. The farm is a medium scale dairy operation on 179.6 ha total farm land, with a total effective area of approximately 175 ha. The farm is currently run as a dairy platform split calving, twice a day milking. Last season 507 Frisian cows were peak milked and produced 605 kgMS/cow. The property sits in the Northern Wairoa surface water catchment and water from the farm flows into the significant Northern Wairoa River which is then eventually received by the Kaipara Harbour.

The System 5 dairy farm currently targets a stocking rate of 2.9 cows/h and incorporates maize, silage, PKE, DDG, Molasses and Canola as supplements, all stored and fed on-farm. According to the NZ Land Inventory NZMS 290 Series, the land is primarily made up of Te Kopuru sand on the majority of the flats with areas of Parore peaty sandy loam in the small gullies. In terms of Land Use Capability class (LUC), the farm ranges from 3w4 to 4e10 with no land in the 7e to 8e range.

This plan assesses the various risk factors to water quality and details agreed actions and practices to be carried out to manage these risks. These are captured in the form of 'good farming practices' (GFPs). Where a GFP is likely being met, this is highlighted in the overview sections of this FEP (see below). Where a GFP is unlikely being met, an open action and timeframe maybe specified.

The farm complies with a high percentage of Good Farming Practices and continues to work towards their environmental goals. By completing this Farm Environment Plan, the farm has shown this commitment to enhancing on-farm sustainability. During the initial farm visit, several opportunities for further environmental improvements were noted - primarily non-permanent waterway fencing. These have all been completed at the time of this review.

OPEN ACTIONS



Continue to keep auditable environmental records

Keep good records of environmental management. This is a valuable habit to maintain for auditing and review process where good environmental practice needs to be demonstrated. This is particularly important for supplementary feed, fertiliser use and location of spreading and stock numbers.

FARM MANAGEMENT



FARM MANAGEMENT

F8 WATER USE OVERVIEW

The dairy shed/yard water is sourced from on farm bore system. This water is used for dairy shed use, plant use and stock water. Rainwater is collected from the dairy shed roof. The stock water is pumped to troughs via alkathene reticulation system and lines are regularly checked for leaks.

A consent for use of water in the farm dairy has been granted by Northland Regional Council for 254.8 cubic meters per week (expires 2051) and a water meter has been fitted to measure water use in the farm dairy.

The farm indicated it has no issues with water sources drying up over the summer months and regularly checks troughs to ensure stock water is topped up.



FARM MANAGEMENT

F12 INFRASTRUCTURE, STORAGE, WASTE OVERVIEW

The farm feeds maize, pasture silage, molasses, DDG, PKE, soya bean hull and canola. The farm has an allocated area with feed storage infrastructure and has appropriate leachate and contaminant capture points that feed into the effluent storage ponds. Maize and pasture silage is stored in several storage bunkers within the feed storage area. Pasture is also bale wrapped and the owners are conscious of wilting the silage during preparation to avoid the effects of leachate. PKE, canola, SBH and DDG is stored several concreted bunkers. These are covered with polywrap. There is no significant waterway near the area.

Only low quantities (1T bags) of Fertilizer is stored on-farm and this is covered. Bulk fertilizer is spread by contractors.

The farm currently disposes of of farm refuse at the local refuse centre. A farm dump is not present. Silage wrap and plastic containers are recycled.

Dead stock are buried on-farm. These burial site(s) are located >50m from any significant water source. NRC permitted activity rules require separation of 50m to water, 50m to water supply bore, 50m to dwelling owned/occupied by another person, 20m to public road. Please adequately cover the burial site to prevent nuisance smells and access by vermin and remove dead animals from streams or watercourses for proper disposal.



Silage stack



PKE bunker



Maize stack



Feed bunkers

FARM MANAGEMENT

F13 FEED PAD

A concrete based feed pad is used on-farm to supply stock with supplements and stand off for stock in severe weather. During peak feeding period, 100% of stock use the infrastructure for feeding 4 months (Sept-Dec) of the year, 7-8 hours a day. 50-80% of stock use the pad for the rest of the year, 5-8 hours a day. All effluent is hosed to an effluent storage pit at the base of the pad then piped to the 1st effluent storage pond.

The farm keeps entry and exits of the pad free from built up effluent by scraping, storing and eventually spreading to land. The use of the pad has helped build a successful farming operation along with reducing their environmental impacts on the land and waterways.



LAND & SOIL MANAGEMENT

WHENUA ME TE ONE



LAND & SOIL MANAGEMENT

GOOD FARMING PRACTICES

Critical Source Areas and farm Hot Spots are identified and managed to minimise contaminant losses to waterways

ACHIEVED

Practices:

Critical Source Areas where there is a risk that contaminants in surface runoff may enter waterways are identified

A grass buffer strip or riparian plantings have been left between waterways and fences

Bridges and culverts have raised sides or mounds to stop runoff entering waterways

If the track or gateway is beside a waterway, the ground is sloped in the opposite direction to avoid run-off

Where possible tracks and gateways are located away from waterways and low points in the landscape

Stock crossing have raised sides or mounds (bundled) to prevent run-off into waterways

Supplement is fed out away from waterways

Troughs are located away from waterways, low points in the landscape or wet areas

Gateways are in a dry point and are wide enough for good cow flow to reduce pugging

Cultivation is managed to reduce the risk of sediment loss and maintain soil structure



ACHIEVED

Practices:

Pugging and compaction of soils is avoided

Erosion-prone land is managed or retired to minimise soil losses



ACHIEVED

Practices:

Planting is done to protect areas from erosion when practical

Areas are permanently fenced off or managed to prevent erosion

Grazing of pastures and crops is managed to minimise sediment and contaminant loss

ACHIEVED

Practices:

If paddocks near waterways are used during wet periods, a buffer strip beside the waterway is fenced off

A larger feeding area is offered in cold conditions when demand is high and utilisation low

LAND & SOIL MANAGEMENT

GOOD FARMING PRACTICES CONTINUED

Paddocks selected for Intensive Winter Grazing (including intensive baleage wintering) are low risk and managed to minimise the risk of erosion, run-off to waterways and leaching to groundwater

ACHIEVED

Practices:

Low risk paddocks are selected for intensive winter grazing that:

- are away from waterways
- are not vulnerable to pugging or compaction
- are 10o or less slope with as few gullies and swales as possible
- are not vulnerable to erosion
- do not have artificial drainage that directly discharge to waterways
- can be easily replanted as soon after grazing as is practical

LAND & SOIL MANAGEMENT

L5

LAND & SOIL OVERVIEW

Northland's climate, topography, historic vegetation and complex geology have combined to form a very complex pattern of soils across the region. Understanding the capabilities and limitations of your soils and Land-use capability is the key to sustainable production in Northland; please refer to the information provided by the Northland Regional Council soil factsheet for farm scale analysis.

The farm currently grows maize for supplementary feed and as part of the wider farm pasture renovation policy. Crop paddocks are tested to determine appropriate fertilizer requirements. Paddock preparation typically includes rip, power harrowed. The cropped paddocks are then re-grassed in perennials. No winter cropping is implemented.

Cultivation at the correct soil moisture levels is always important to protect soil structure and productivity.

Under NRC's current cultivation rules, land preparation (cultivation, excluding direct drilling) is not allowed within 10m of Inanga spawning sites, lakes and natural wetlands and continually or intermittently flowing rivers. These buffer zones intercept and filter runoff from cultivated land, keeping sediment out of the waterways.

The cultivation setback from rivers can be reduced to 5m if the slope of the land adjoining the riverbed is 10 degrees or less and appropriate sediment control measures are installed.

The farm races are well structured and maintained with brown rock. At the time of visit, the likelihood of sediment, phosphorus and microbial pathogens being lost to surface water via the races was fairly low risk throughout the farm due to the low lying nature of the land.

The farm utilizes a several land management strategies to minimize pugging, pasture damage and soil loss on farm.

These include:

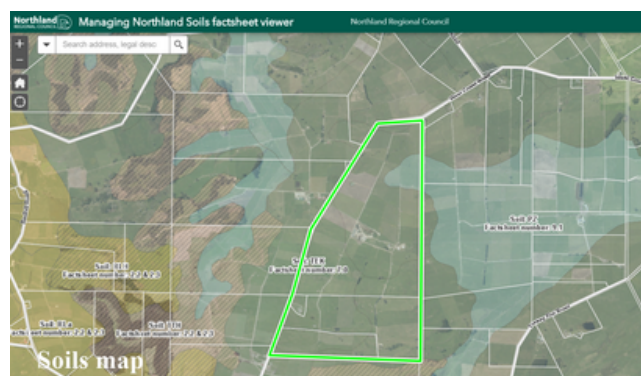
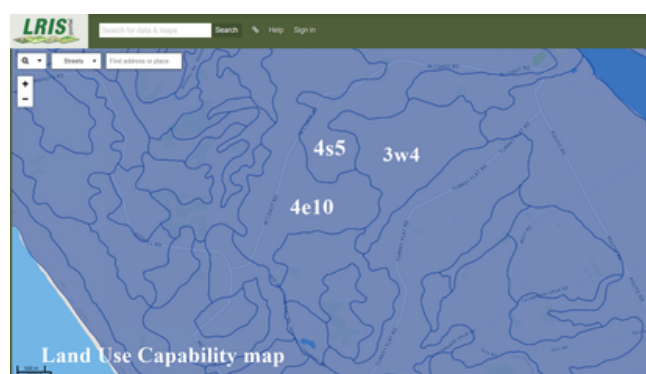
- Medium stocking rate (<3 cows per ha);
- On / off grazing with use of covered feed pad
- Farm drains and waterways are permanently and temporarily fenced to exclude stock;
- Selective paddock selection in wet weather including the use of a wintering (rock) paddock;
- Stock well fed to minimize pugging in paddock;

LAND & SOIL MANAGEMENT

L7 SOIL TYPES & LAND USE CAPABILITY

The soils on the dairy platform are primarily made up of Te Kopuru sand and a sections of Parore peaty sandy loam – please see soil map and refer to factsheet.

In terms of Land use capability (LUC), the farms 3 main classes are 3w4 – 4s5 – 4e10 range. The majority of this land is consistent with old sand terraces and a small section of peatland. No retirement land is captured under the governments LUC layer on the farm.



WATER USE & IRRIGATION MANAGEMENT WHAKAMĀKŪKŪ



WATER USE & IRRIGATION MANAGEMENT

GOOD FARMING PRACTICES

Dairy shed and stock water use is efficient and prevents source contamination

ACHIEVED

The depth, rate and timing of irrigation is managed to meet plant demand and minimise the risk of leaching and run-off



N/A

Practices:

Irrigation is done to replace soil moisture deficit only (the soils full-point and irrigation trigger point are correctly set and monitored)

Soil moisture levels and weather when scheduling irrigation are assessed by:

- Estimating soil moisture levels with a soil water budget or
- Monitoring soil moisture levels with real time soil moisture equipment

Irrigation events are recorded - when, where, amount

The irrigation system is designed, operated and regularly checked to minimise the amount of water applied to meet plant demand, and prevent microbial contamination

N/A

Practices:

All new irrigation systems are designed and installed in accordance with industry codes of practice and standards

All irrigation water takes are telemetered

Routine application depth tests are carried out to assess performance

All staff are trained to use the system (Consider Irrigation NZ's operator and manager training)

All well heads are sealed, and stock permanently excluded from them

A backflow prevention system is installed (where required)

A 'wet' commissioning test is undertaken that includes pressure, flow, depth, and uniformity

Pre-season checks, including a visual inspection of the irrigator when it is running, are undertaken for each irrigation system annually

A performance assessment is undertaken for each irrigation system every 5-years or:

- when a major system change occurs
- there is significant non-compliance

EFFLUENT MANAGEMENT PARAKAINGAKI



EFFLUENT MANAGEMENT

GOOD FARMING PRACTICES

Effluent and manure is applied at depths, rates and amounts that match plant requirements and minimise the loss of nutrients or microbial pathogens to waterways and groundwater



ACHIEVED

Practices:

Nutrient load is spread evenly across the largest area practical

Soil tests are taken biennially in the effluent application area, and fertiliser applications adjusted accordingly

Odour impact is considered during application

The effluent system is designed, operated and regularly checked to minimise the risk of nutrient and microbial pathogen loss to waterways and groundwater, and to prevent microbial contamination



ACHIEVED

Practices:

The system design is appropriate for the soil type, topography, and climate

Storage facilities are sealed

Effluent solids that accumulate are routinely removed

Effluent system checks are undertaken at least annually, including testing of effluent depth Effluent equipment is inspected and maintained regularly as and when required

Effluent pumping equipment is routinely serviced

EFFLUENT MANAGEMENT

E2 EFFLUENT OVERVIEW

Effluent is collected and managed at multiple sources, being the dairy shed, dairy yard and milk collection point then gravity fed to effluent storage ponds. As mentioned, effluent from the feed pad is also collected via a drain sump then gravity fed to the 1st effluent pond.

A storage calculation has been completed using the dairy effluent storage calculator (DESC). The calculation indicates that existing storage is sufficient for the current dairy operation (see appendices for DESC report) and the farm does not discharge effluent to waterways.

The farm has good effluent management practices in place and regularly utilizes stormwater water diversion.

The ponds are set well away from neighbouring homes and the risk of odour issues is considered negligible. The ponds are considered to meet the current sealing standard required by NRC and no visual signs of leaking was observed.

At the time of visit, stormwater diversion system was in place. It is essential that good management practice is followed to optimise the performance of the effluent treatment system. The owners are aware to be effective, the use of the stormwater diversion system must be part of the daily routine and is extremely important to manage if pressure is on storage capacity.



1st Effluent storage pond (solids)



Effluent sand trap/solids trap



Diversion from concrete pad



2nd Effluent storage pond

EFFLUENT MANAGEMENT



EFFLUENT MANAGEMENT

E6

EFFLUENT IRRIGATION

When soils are in moisture deficit, effluent is irrigated to the flats using a travelling irrigator. The effluent irrigation area is located centrally on the farm and comprises approximately 60 ha of pasture. This is considered large enough to minimise the risks of K loading.

The soil in these areas are generally imperfectly drained and, as the majority of this land is flat, ephemeral flow paths are common. A good practice to incorporate is creating good buffers from these flows paths to ensure irrigation is not lost via surface flow. It is important to understand that where subsurface drainage and ephemeral flow paths are as these provide direct route for excess irrigated effluent to reach waterways if soil uptake capacity is exceeded. The lowland sections can get extremely wet during winter/early spring when rainfall is high therefore opportunities to irrigate during this period are limited. The farm policy is to maintain 20m buffers from waterways.

The farm staff regularly conducts maintenance on the effluent irrigation system including checking reticulation for leaks and conducting a maintenance routine on the pods/irrigator as and when required.

Irrigation method

Travelling irrigator



WATERWAYS & BIODIVERSITY MANAGEMENT

RARENGA RAUROI



Priority 1 - Permanently fence waterway



Priority 2 - Permanently fence + wetland plant

WATERWAYS & BIODIVERSITY MANAGEMENT

GOOD FARMING PRACTICES

Stock are excluded from lakes and waterways

ACHIEVED

Practices:

All stock crossings are bridged or culverted

Drains are well managed

Farm indigenous biodiversity and Mahinga Kai values are identified and protected



ACHIEVED

Practices:

Areas are identified on the farm map

Stock are fenced out of the area

Weeds are controlled within the area

Animal pests are trapped or poisoned

WATERWAYS & BIODIVERSITY MANAGEMENT

W3

WATERWAYS & BIODIVERSITY OVERVIEW

A significant amount of waterway fencing has been completed on the property including all fencing required on permanent flowing waterways along with many ephemeral waterways/drains. The farm incorporates temporary fencing on all non-permanent flowing waterways.

All significant crossings are culverted ensuring stock have no direct access to waterways and are of good quality and well structured. No perched or block culverts were observed on farm that may disturb fish passage.

Biosecurity policies are implemented when require however the farm does not use a biosecurity management plan. Kauri's are not present on farm therefore Kauri dieback mitigation does not need to be considered at this time.

Pests are not generally a problem on the surrounding land as bush/forests are reasonably scarce. Pest control is implemented on-farm. Bait stations for rats, and there are loads of stations around effluent ponds, feed pad, sheds and houses. They are monitored and refilled weekly, all year. Possum traps are also in use and checked regularly. If further pest control is an interest, there is potential opportunity to contact the NRC for assistance, or there is comprehensive information available online through <https://www.nrc.govt.nz/environment/weed-and-pest-control/pest-control-hub/>. Weed control is regularly implemented on various pasture weed species.



WATERWAYS & BIODIVERSITY MANAGEMENT



PRIORITY 1 - PERMANENTLY FENCE WATERWAY

UPDATE: This area has been fenced.

This plan has identified these waterway seepages as areas that would benefit from permanent fencing and eventually riparian planting. A 1 or 2 wire electric would be sufficient.

Excluding stock and retiring this land will improve the functioning of the waterway by reducing the amounts of nitrogen, bacteria and sediment reaching waterways.

WATERWAYS & BIODIVERSITY MANAGEMENT

W11 PRIORITY 2 - PERMANENTLY FENCE + WETLAND PLANT

UPDATE: This area has been fenced.

This wetland area would benefit greatly with permanent fencing and wetland planting. This area would be an excellent example of retiring unproductive land and enhancing wet habitats by improving water quality through filtration and shading.

The area would be ideal to plant in natives over a season. May to August is the best time to plant. The site would require approximately 500 native riparian plants including;

- Cabbage trees - On the edges- Flax (Harakeke) – On the edges- Manuka – On the drier edges



NUTRIENT MANAGEMENT TAIORA



NUTRIENT MANAGEMENT

GOOD FARMING PRACTICES

Soil phosphorus levels are monitored and maintained below or within the target ranges for the soil-type and crop



ACHIEVED

The amount and timing of fertiliser inputs, takes account of all sources of nitrogen and phosphorus, matches plant requirements and minimises losses to waterways and groundwater



ACHIEVED

Practices:

All fertiliser applications are recorded – product, rate, date, location (If a contractor is used the information is gathered from them)

Soil temperature and moisture levels are assessed before applying fertiliser (i.e. avoid winter months)

Pasture or crop growth rates are considered along with feed requirements before applying N

Fertiliser spreading equipment is maintained and calibrated



ACHIEVED

Practices:

Farm spreading equipment is calibrated regularly specific to the product being spread -- spreading width and volume checked

Paddocks are checked for paddock stripes after spreading

Contractors are Spreadmark accredited

NUTRIENT MANAGEMENT

N1 NUTRIENT OVERVIEW

Effective nutrient management is a win:win on farm – maximising pasture and crop production and minimising loss of nutrients to waterways and the atmosphere. Understanding the current levels of nutrients in the soil, and their trends is critical to this.

While sediment and faecal contamination are the primary contaminants of waterways in Northland, phosphate is also a contaminant for consideration. As the source of P in waterways is primarily sedimentation, farmers are encouraged to maintain their soil fertility values (Olsen P) to be in the optimum range for pasture production (these are generally 25-40).

Where P is above optimum P loss associated with any loss of soil (eg from pugging) will be significant, and the risk of this exceeds any agronomic benefit from increasing availability of P to pasture.

Soil tests are carried out at least every second year to support nutrient management decisions, this includes testing of the effluent block. Deep Soil tests (to 150mm for maize) are also carried out prior to any cropping events.

The farm participates in Fonterra's nitrogen programme each season by submitting information on farm inputs / outputs and management decisions. Nitrogen use last season was higher than the regional average at 188 kgN/ha - but Home grown feed eaten (11.1 T DM/ha) was in the top 25% for the region.

N-fertilizer is applied using on-farm equipment and helicopter contractors as appropriate. Proof of placement is implemented when applying fertiliser. Farm staff use a farm map to record placement, and contractors provide the farm with Tracmap.

Archived GMP

General nutrient management

Practices:

Soil-testing is done each year for each different management block
Soil-testing is done well before crops are planted to identify nutrient levels

A nutrient budget is used to help fertiliser decision making
Supply farm nutrient information to your milk company at the end of each season

Evidence:

Soil test results
Nutrient budget
End of season nutrient budget
Nutrient management plan

GREENHOUSE GAS EMISSIONS

TUKUNGA HAU KŌTUHI



GREENHOUSE GAS EMISSIONS

GOOD FARMING PRACTICES

Farm greenhouse gas emissions are known and a plan is in place to reduce or offset them, that also considers adaptation to climate change

Practices:

Greenhouse gas emissions are calculated annually for the farm, and opportunities to improve on-farm efficiency are explored.



ACHIEVED

WHAKAPAPA



APPENDIX

THANK YOU

DISCLAIMER

*Provision of advice in relation to effluent storage, effluent irrigation systems and the management of other environmental risk areas on farm.

The advice that Fonterra Co-operative Group Ltd (Fonterra, we, us) provides to farmers in relation to effluent storage capacity and other environmental compliance practices, including mitigation actions described in Farm Environment Plans, is based on the information and assumptions that farmers and their agents have provided to us and on our knowledge and understanding of current best practice in the industry. Fonterra does not purport to replace sound engineering or other professional advice and as such we strongly encourage farmers to seek independent expert advice before any construction, upgrades, or other change to your on-farm practices. Farmers are ultimately responsible for the environmental compliance of their farm and on-farm practices. Fonterra gives no warranties (express or implied) and, to the maximum extent permissible by law, excludes all liability in contract or tort (including, without limitation, liability for negligence) or otherwise in relation to the advice provided.